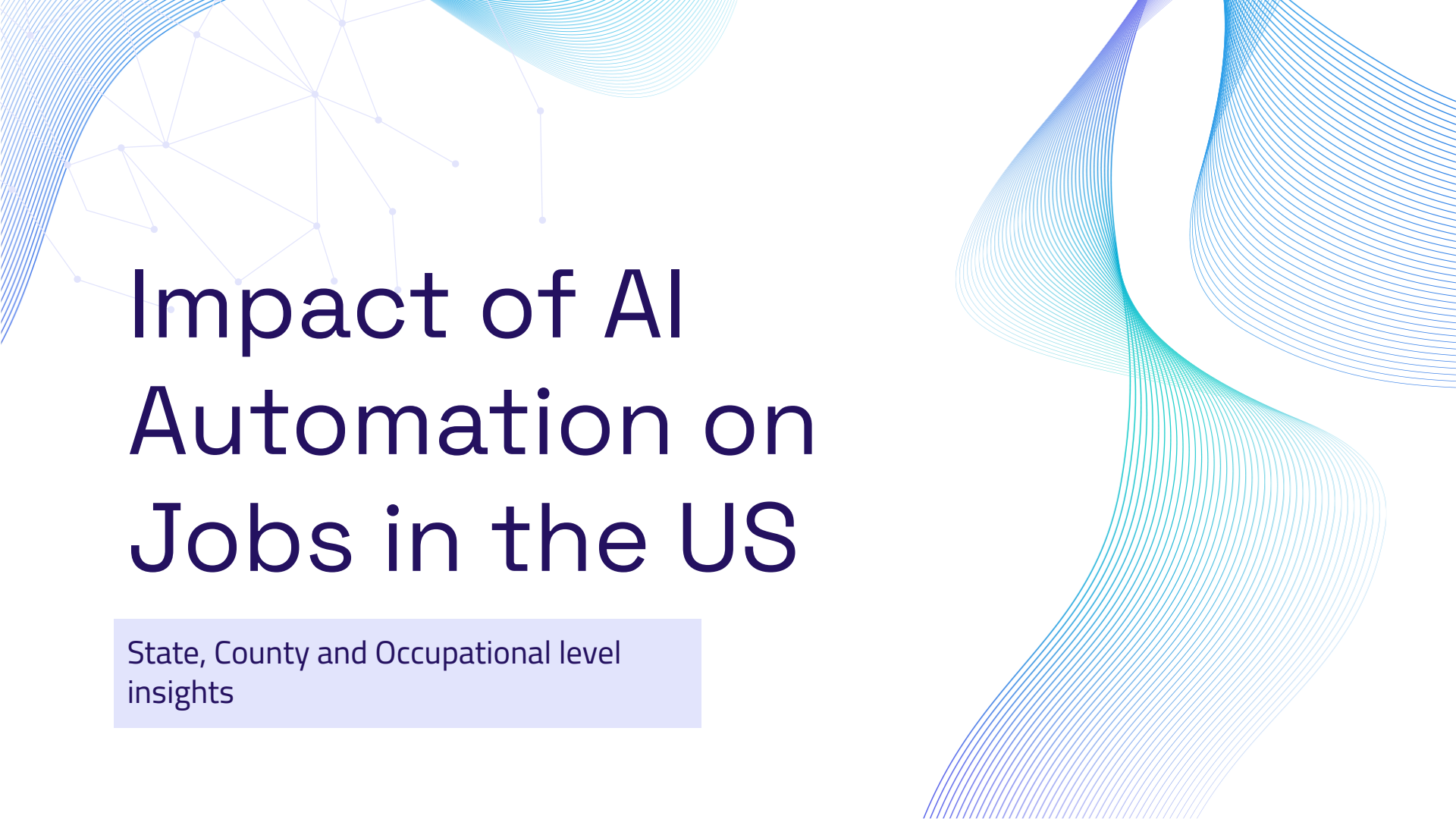




Impact of AI Automation on Jobs in the US

State, County and Occupational level
insights

Project Introduction



Artificial intelligence is rapidly transforming the U.S. labor market, reshaping how work is performed across industries and regions. As automation technologies become more capable and affordable, a growing share of routine and repetitive tasks can be executed by automation backed systems, increasing the risk of job displacement and consequently economic impact for the government. Understanding where this risk is concentrated is critical for policymakers, businesses, and workers preparing for the next wave of technological change.

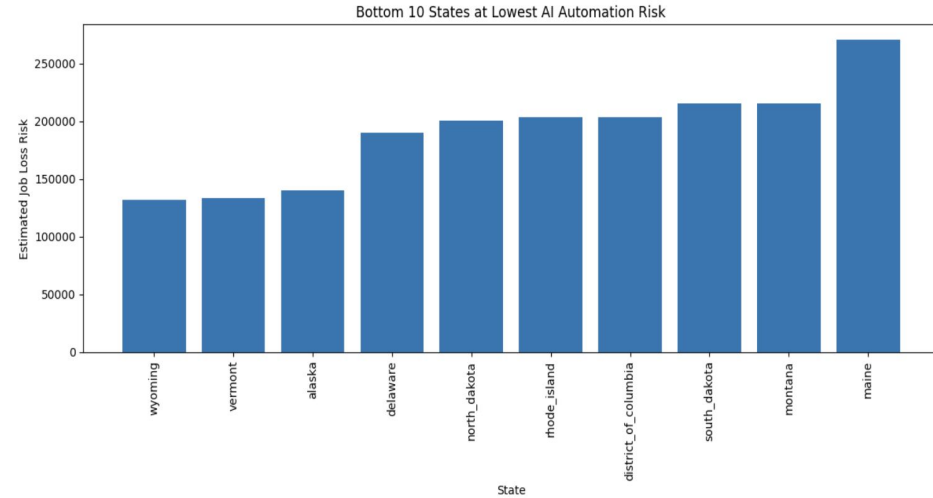
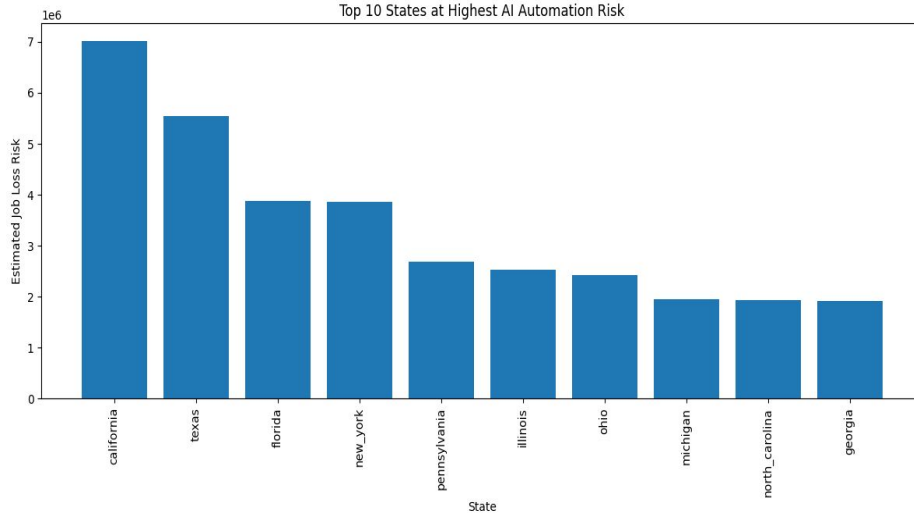


This project analyzes the geographic and occupational distribution of AI automation risk across the United States, and takes a deep dive in the state of California. It examines which states and counties face the greatest exposure, identifies the occupations most vulnerable to job loss, and evaluates how task-level automation probabilities translate into real-world labor impacts. By combining automation probability scores with employment and regional population data, this study provides a comprehensive view of what repercussions the US labor force might face and points out where economic consequences are likely to be the most severe.

Data Source

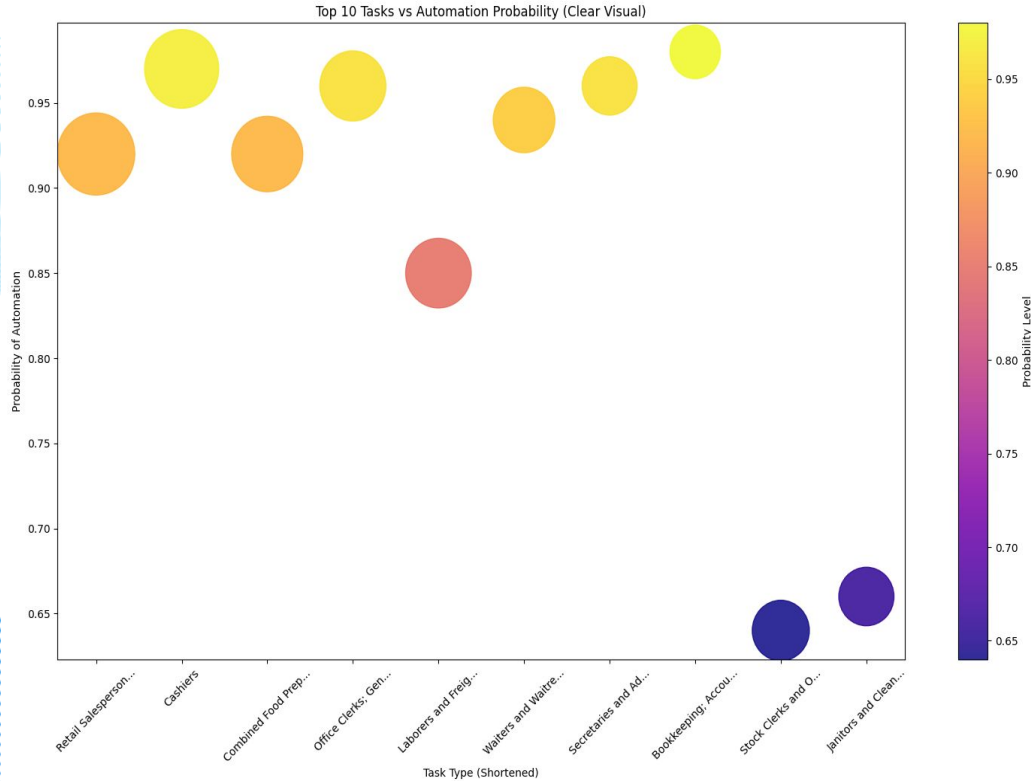
Dataset	Type	Format	Raw Data Size
O*NET Database	File	CSV	18798
ai_job_trends.csv	API	HTML response -> JSON -> CSV	10006
county_data_1	File	CSV	67
Frey & Osborne(2017) Automation Risk	File	CSV	703

State Level Automation Exposure



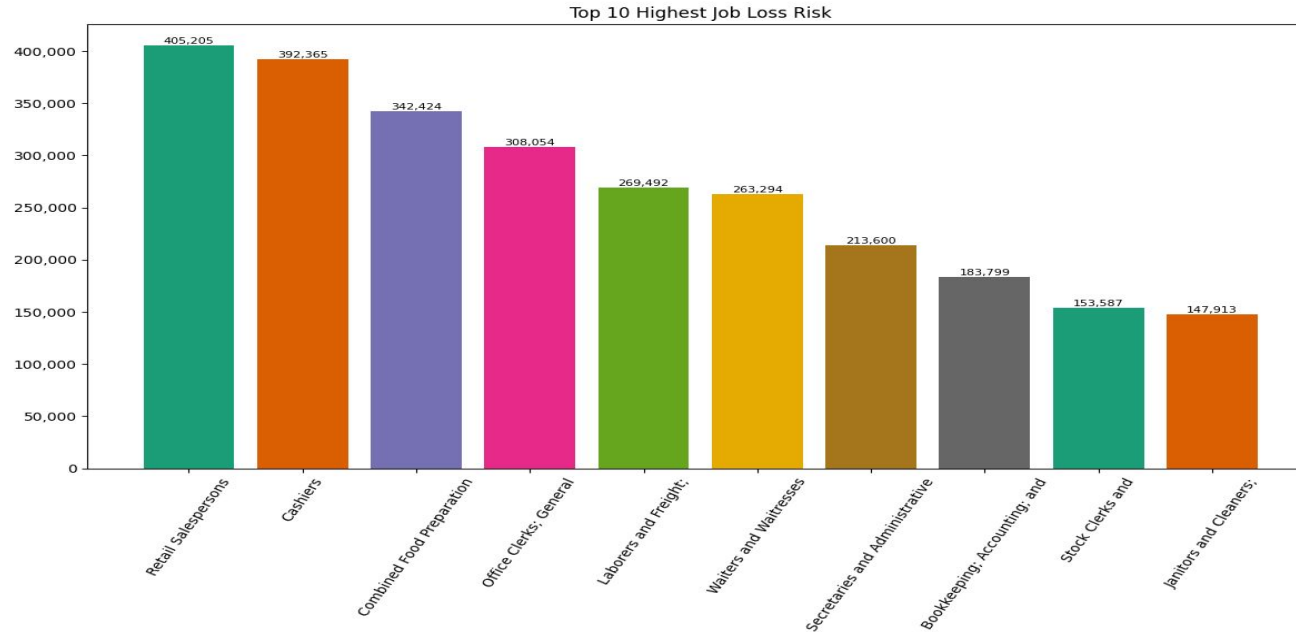
- States like California, FLorida, Texas and New York show absolute numbers of approximately 4-7 millions jobs at risk.
- The risk is primarily driven by the size of the population economies which a large proportion of service sectors presence in them.
- States like Wyoming show lower jobs at risk because of small population effectively small working labor force
- Jobs in similar sectors i.e. service sectors are at risk even in states facing the lowest risk.

Task Level Automation Risk



- Tasks like retail salesperson, cashiers, office clerks, laborers face extremely high risk more than 90% of being automated.
- One of the major contributing reason towards this risk is the redundancy and repetitive nature of the task.
- High concentration of these job roles containing these tasks in states like California make it prone to such high risk.

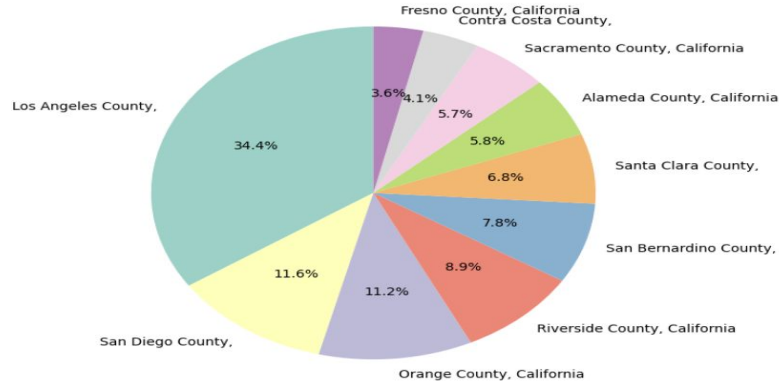
Occupation Level Job Loss



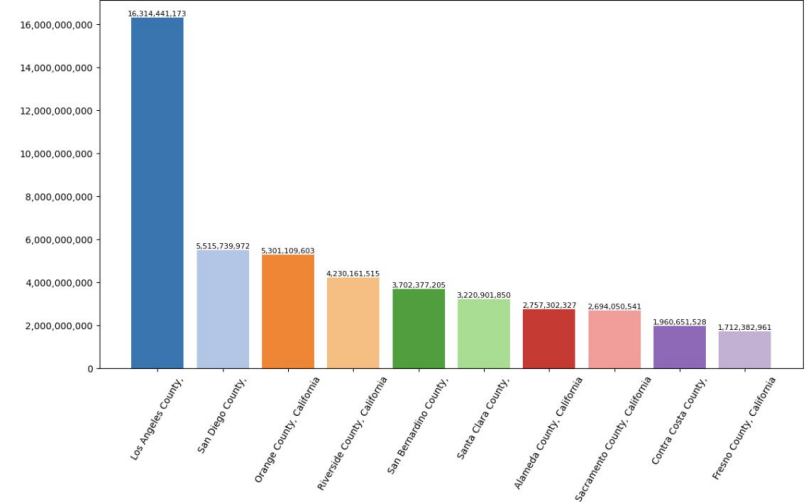
- Retail associates, cashiers and clerks face the highest risk of being replaced by AI
- The problem is that these roles no matter supplemental in nature are present across industries therefore this graph shows that the this automation cycle will impact almost all industries and mass population
- The primary reason for job loss is the sheer number of people working in these roles and by the nature of the work, as reducing the workforce can help in cost cutting for employers

County Cost Impact

Top 10 Counties by Share of California Population



Top 10 Counties by Automation Cost Impact



- Automation cost of the entire state is driven by few counties fundamentally
- Los Angeles County having the largest share of population in CA alone accounts for about 40-45% of impact California.
- Counties like LA face such disproportionate economic impact because of presence of large number of service sector jobs

Challenges

- One of the main challenges was merging different datasets that used different naming conventions and levels of granularity. County names in particular required cleaning because formats weren't consistent across sources.
- Another challenge was mapping occupation automation probabilities to employment counts in a reliable way.
- Interpreting task and occupational exposure required clear separation of probability and numerical impact metrics.
- Finally, because exposure varies so much across states and counties, creating visualizations that clearly showed differences without distorting the results required formatting.